Lesson 2.3

Postulates & Diagrams

Rules we accept — and what a picture really tells you

What is a postulate?

A **postulate** is a rule we accept as true **without proof**.
Everything else in geometry is built on postulates.

Point, line and plane postulates



Through any **two points** there is exactly **one line**.

Point, line and plane postulates

Through any **two points** there is exactly **one line**.



Through any **three noncollinear points** there is exactly **one plane**.

Point, line and plane postulates



Through any **two points** there is exactly **one line**.



Through any **three noncollinear points** there is exactly **one plane**.



Two lines meet in exactly **one point**.

Point, line and plane postulates



Through any **two points** there is exactly **one line**.



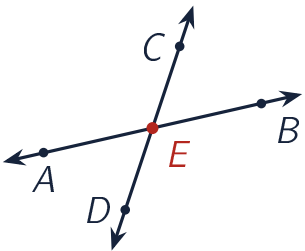
Through any **three noncollinear points** there is exactly **one plane**.



Two lines meet in exactly **one point**.

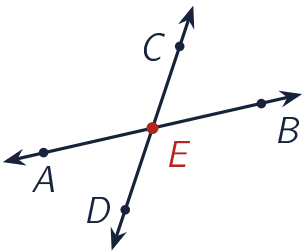
Two planes meet in exactly **one line**.

Reading a diagram



You **can** assume: A, E, B are collinear ✓

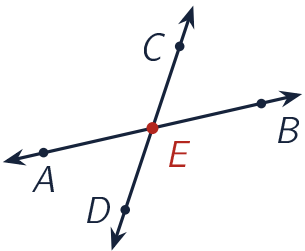
Reading a diagram



You **can** assume: A, E, B are collinear ✓

You **can** assume: the two lines meet at E ✓

Reading a diagram

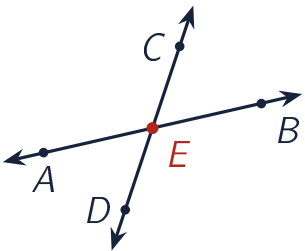


You **can** assume: A, E, B are collinear ✓

You **can** assume: the two lines meet at E ✓

You **cannot** assume: $\angle AEC = 90°$ × (no right-angle mark)

Reading a diagram



You **can** assume: A, E, B are collinear ✓

You **can** assume: the two lines meet at E ✓

You **cannot** assume: $\angle AEC = 90°$ × (no right-angle mark)

You **cannot** assume: $AE = EB$ × (no tick marks)

Marks tell you sizes


right angle


equal lengths

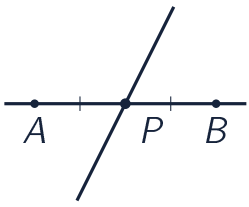

parallel

Watch out

Watch out! "It looks like it" is **not** a reason. Only what is **drawn** (position) or **marked** (size) counts.

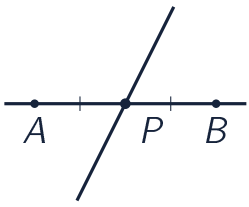
Your turn! Pause the video and try it.

Which can you assume?
(a) A, P, B are collinear (b) $PA = PB$ (c) the lines are perpendicular



Your turn! Pause the video and try it.

Which can you assume?
(a) A, P, B are collinear (b) $PA = PB$ (c) the lines are perpendicular



(a) yes — drawn on one line. (b) yes — tick marks. (c) no — no right-angle mark.

Remember

- Two points $\rightarrow$ one line; three noncollinear points $\rightarrow$ one plane.
- Two lines meet in a point; two planes meet in a line.
- Assume what is **drawn**, not sizes — unless they are **marked**.

Now practice at your level:

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